Linear Approximation and L'Hôpital's Rule

ANSWER KEY

Tangent line approximation

1 Use linear approximation to estimate $\sqrt{4.1}$, given $f(x) = \sqrt{x}$, $f(4) = 2$, and $f'(4) = \frac{1}{4}$.
$L(x) = f(4) + f'(4)(x-4) = 2 + \frac{1}{4}(x-4)$. $L(4.1) = 2 + \frac{1}{4}(0.1) = 2.025$.

2 Use linear approximation to estimate $(1.02)^5$, using $f(x) = x^5$ near $x = 1$.
$f(1) = 1$, $f'(x) = 5x^4$, $f'(1) = 5$. $L(x) = 1 + 5(x-1)$. $L(1.02) = 1 + 5(0.02) = 1.1$.

3 Find the linearization $L(x)$ of $f(x) = \sin x$ at $x = 0$, and use it to approximate $\sin(0.1)$.
$f(0) = 0$, $f'(0) = \cos 0 = 1$: $L(x) = x$. $\sin(0.1) \approx L(0.1) = 0.1$.

Error and over/underestimation

4 For $f(x) = \sqrt{x}$ (concave down for $x > 0$), explain whether the linear approximation at $x = 4$ overestimates or underestimates $f(4.1)$.

Since f is concave down, its graph lies below any tangent line, so the tangent line (linear approximation) OVERESTIMATES the true value of $f(4.1)$.

5 For $f(x) = x^2$ (concave up everywhere), explain whether the tangent line approximation at any point overestimates or underestimates nearby function values.

Since f is concave up, its graph lies above any tangent line, so the tangent line approximation UNDERESTIMATES nearby true function values.

Indeterminate forms and L'Hôpital's Rule

6 Evaluate $\lim_{x\to 0} \frac{\sin x}{x}$ using L'Hôpital's Rule, and confirm it matches the known limit.
Form $\frac{0}{0}$: differentiate top and bottom: $\lim_{x\to 0} \frac{\cos x}{1} = \cos 0 = 1$, matching the well-known limit.

7 Evaluate $\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$ using L'Hôpital's Rule (applied twice).
Form $\frac{0}{0}$: $\lim_{x\to 0} \frac{e^x - 1}{2x}$, still $\frac{0}{0}$: apply again: $\lim_{x\to 0} \frac{e^x}{2} = \frac{1}{2}$.

8 Evaluate $\lim_{x\to\infty} \frac{\ln x}{x}$ using L'Hôpital's Rule.
Form $\frac{\infty}{\infty}$: $\lim_{x\to\infty} \frac{1/x}{1} = \lim_{x\to\infty} \frac{1}{x} = 0$.

Recognizing when L'Hôpital's Rule applies

9 Explain why L'Hôpital's Rule cannot be directly applied to $\lim_{x\to 0}\frac{x+1}{x}$, and evaluate the limit by other means.

This is NOT an indeterminate form — as $x \to 0$, the numerator approaches 1 (not 0) while the denominator approaches 0, so it's a $\frac{1}{0}$ form, not $\frac{0}{0}$. This limit does not exist (it approaches $\pm\infty$ depending on direction), so L'Hôpital's Rule doesn't apply.

Differentials

10 For $f(x) = x^3$, find the differential dy when $x = 2$ and $dx = 0.1$, and compare it to the actual change $\Delta y = f(2.1) - f(2)$.

$dy = f'(x)\,dx = 3x^2\,dx = 3(4)(0.1) = 1.2$. Actual: $f(2.1) = 9.261$, $f(2) = 8$, $\Delta y = 1.261$. The differential $dy = 1.2$ closely approximates the actual change $\Delta y = 1.261$.

More L'Hôpital's Rule practice

11 Evaluate $\lim_{x\to 0}\frac{\tan x - x}{x^3}$ using L'Hôpital's Rule (applied as many times as needed).

Form $\frac{0}{0}$: $\lim_{x\to 0}\frac{\sec^2 x - 1}{3x^2}$, still $\frac{0}{0}$: apply again: $\lim_{x\to 0}\frac{2\sec^2 x\tan x}{6x}$, still $\frac{0}{0}$: apply once more: $\lim_{x\to 0}\frac{2\sec^2 x(\sec^2 x) + 4\sec^2 x\tan^2 x}{6} = \frac{2(1)+0}{6} = \frac{1}{3}$.